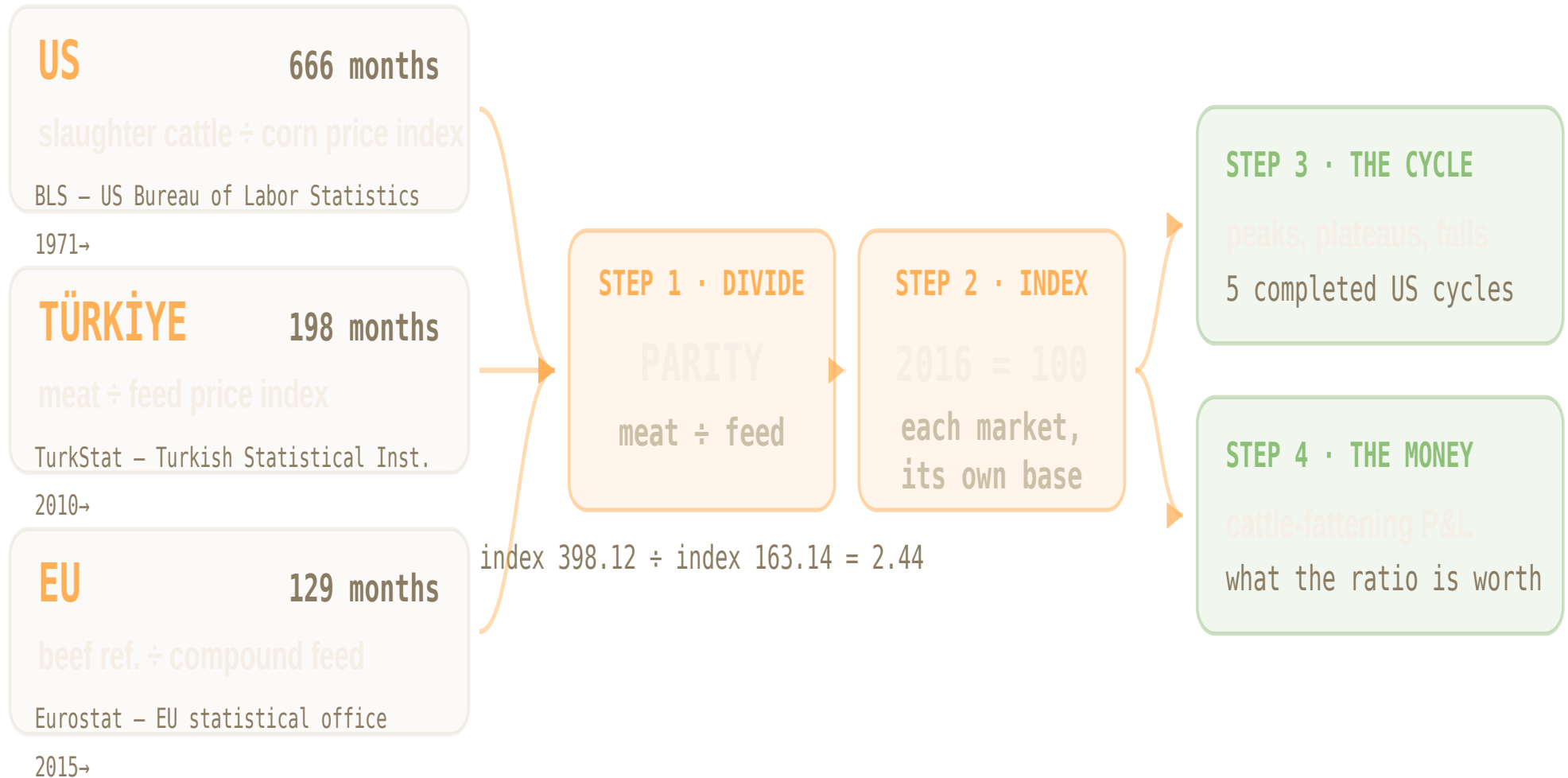


55 years of US data. 16 of Türkiye. 11 of the EU.

Six published price series — every monthly observation each country has — consolidated into one comparable measure of what a livestock producer earns, then read for its cycle.



Cattle producers in all three of our markets are at a cycle peak.

+84%_{US} **+25%**_{EU} **+21%**_{TR}

more feed a kilo of beef buys today than it did in 2016 — each market against its own past

13

months the US has already spent within 10% of its peak

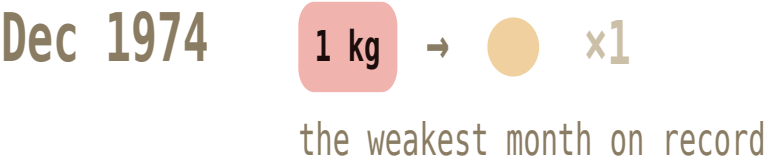
0

of the five previous US peaks that lasted — every one fell back

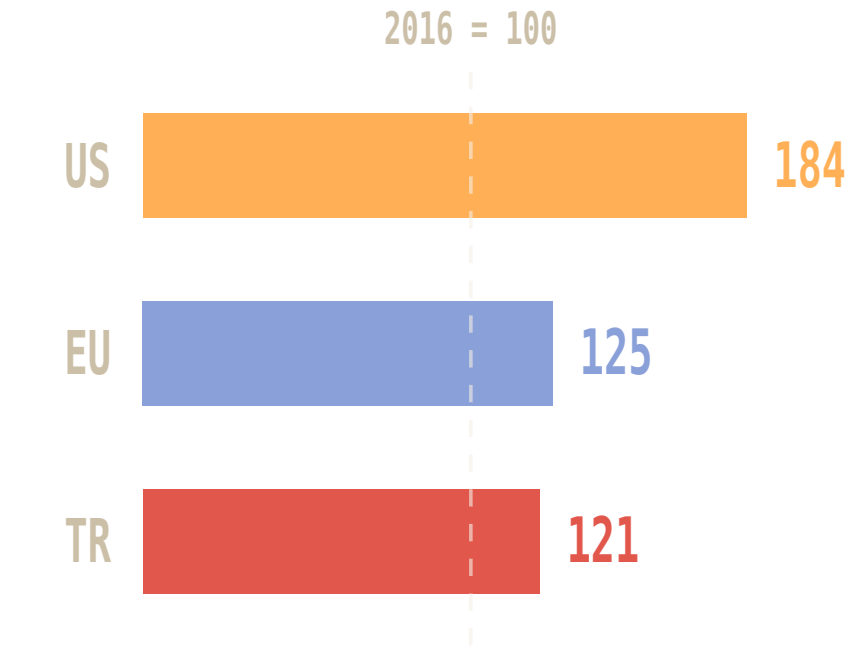
One ratio: what a kilo of beef is worth in feed.

Higher = meat outpacing feed, the producer earns more. It is a relative measure — read it against its own past, never as a quantity of feed.

FEED-BUYING POWER OF A KILO OF US BEEF



ALL THREE MARKETS • EACH VS ITS OWN 2016



US 184 = 84% more feed per kilo of beef than in 2016.

each oval = the Dec 1974 low • ratio of US cattle and corn production index is comparable; raw price levels are

US slaughter cattle ÷ corn

BLS (US federal statistics office) • 1971→ • 666 mo

EU beef ref. price ÷ compound feed

Eurostat • 2015→ • 129 mo

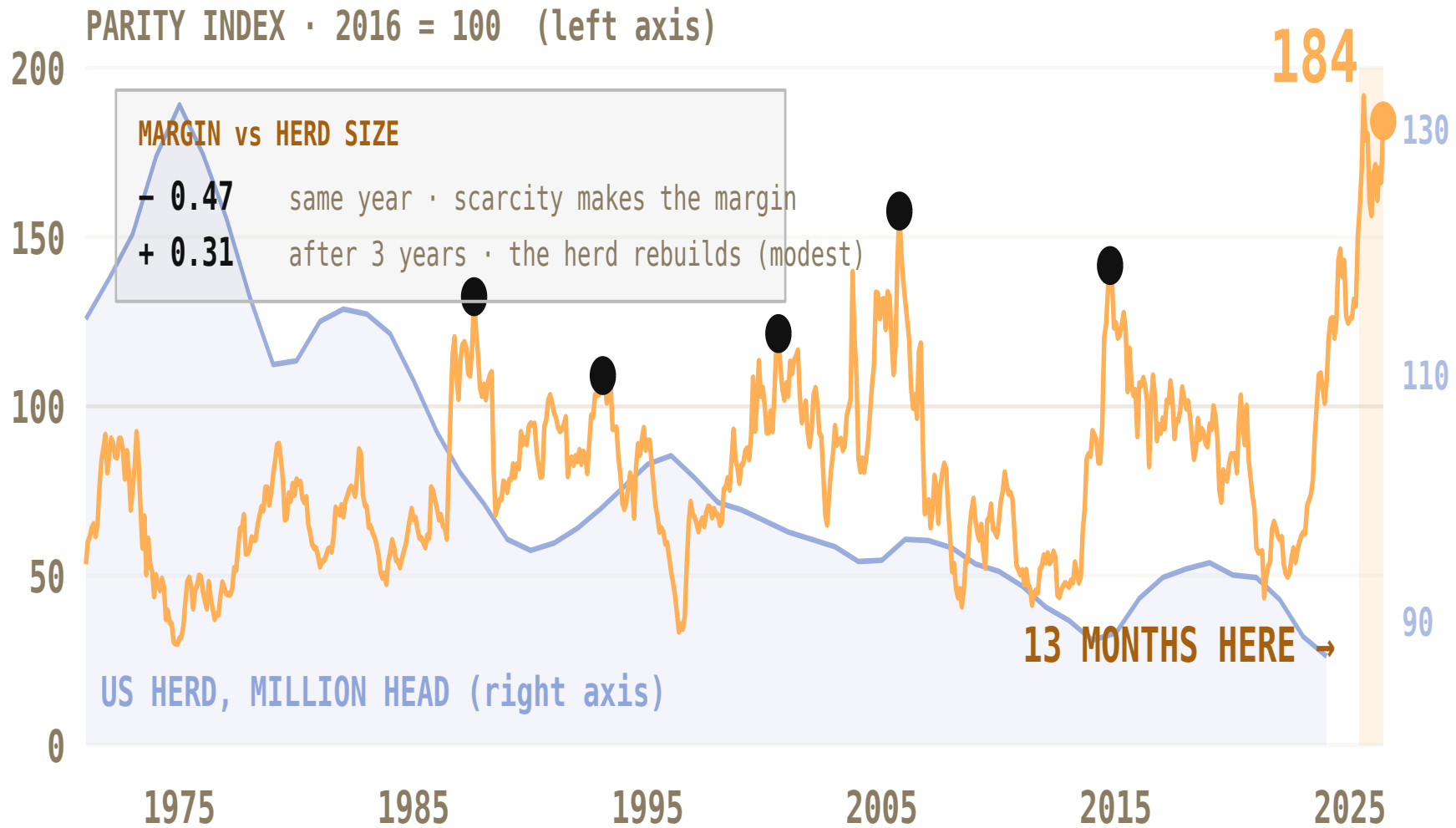
TR meat ÷ feed PPI

TurkStat • 2010→ • 198 mo

Not the same basket in each country — which is why each series is indexed to its own base, and why we read shape rather than level.

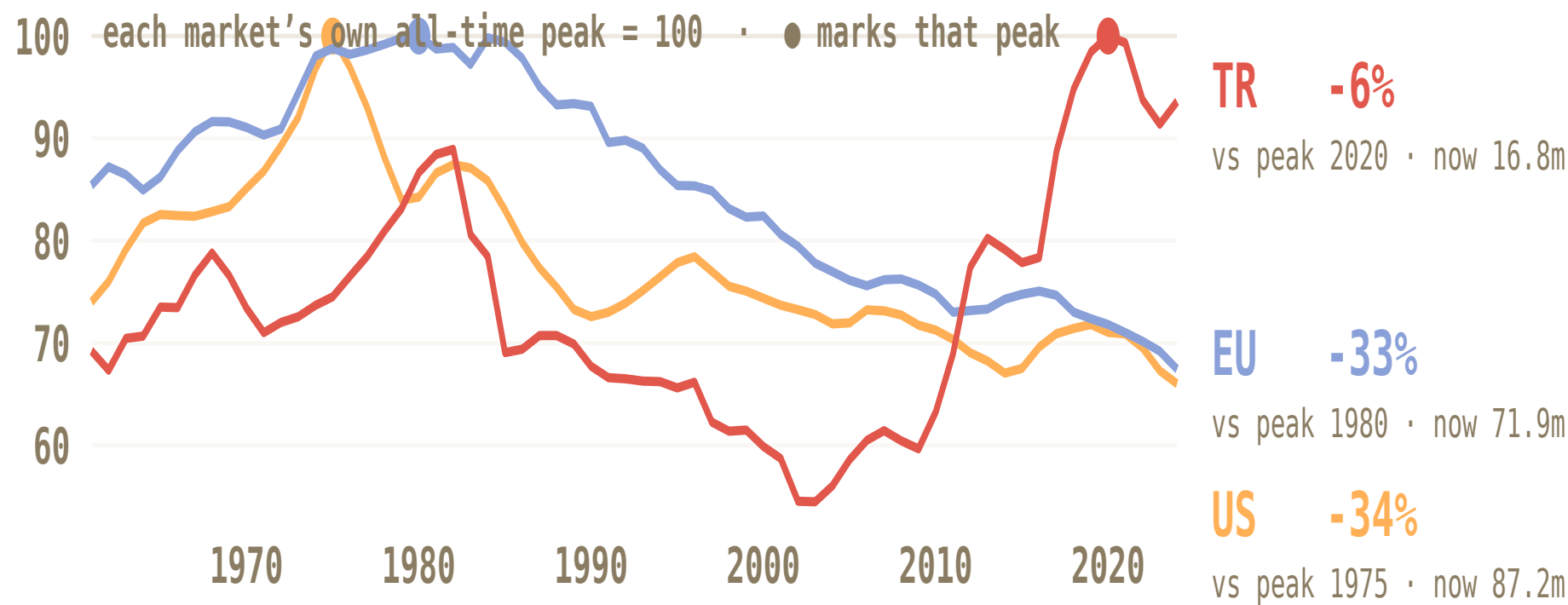
Record margins. Fewest cattle since the series began.

US only. Amber = producer margin ratio (left). Blue = the size of the US cattle herd (right). Scarcity is what makes the margin.



Two herds down a third. One still growing.

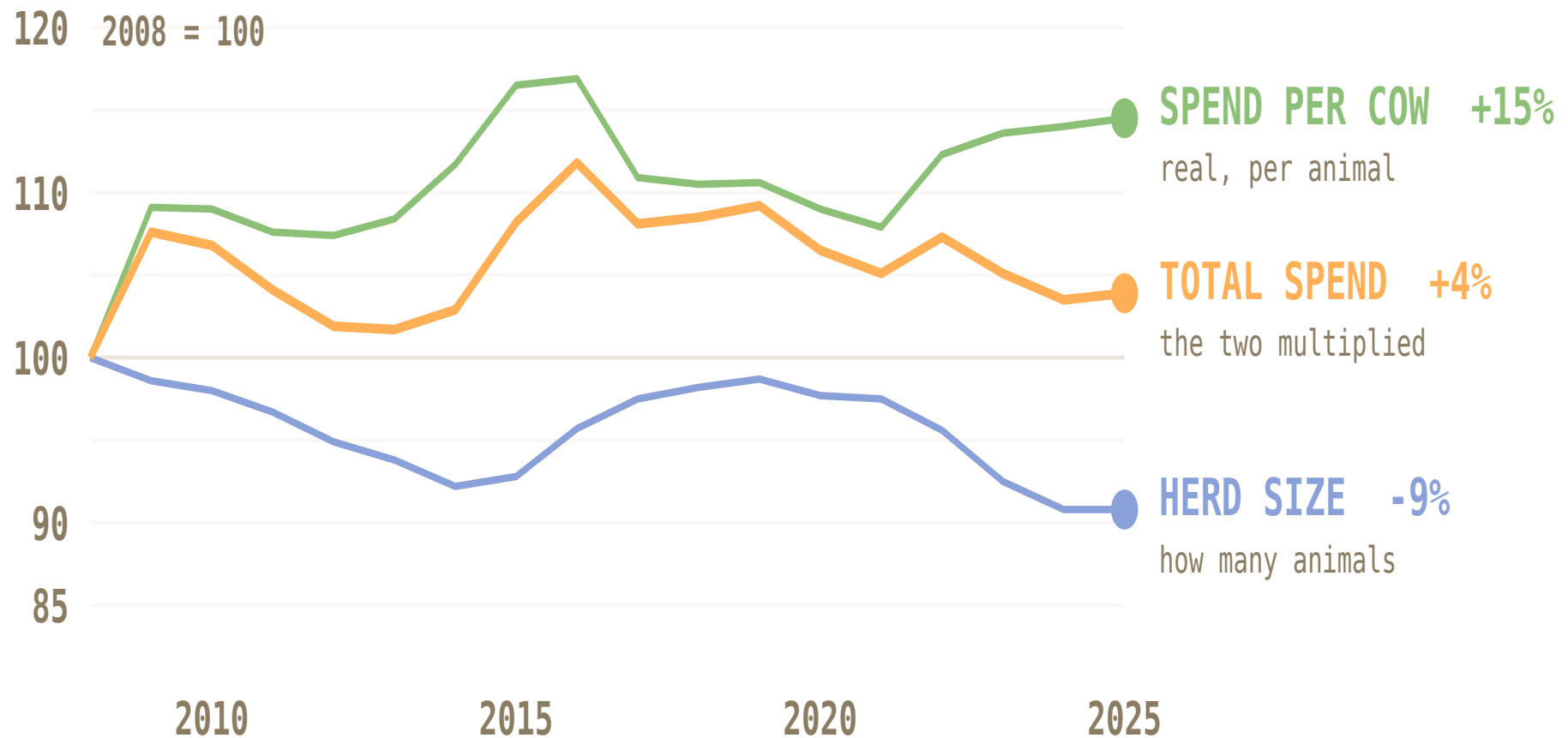
Cattle numbers, each market against its own all-time peak. Margin is measured per animal — this is how many animals there are.



These counts include dairy and beef animals together — FAOSTAT reports all cattle. Parity, by contrast, is a beef measure throughout. That matters most for the EU, where a large part of the decline is dairy restructuring (fewer, higher-yielding cows, accelerating after quotas ended in 2015) rather than the beef cycle. The US herd is predominantly beef, so its line is the closest match to the parity line. Series ends 2024; FAOSTAT publishes with roughly an 18-month lag.

More per animal. Fewer animals. Total flat.

US cow-calf vet & medicine spending, inflation-adjusted. Per animal it rises with the margin ($r = +0.70$) — the shrinking herd cancels it out.



Five US peaks. Every one fell 25–60% — over two to five years.

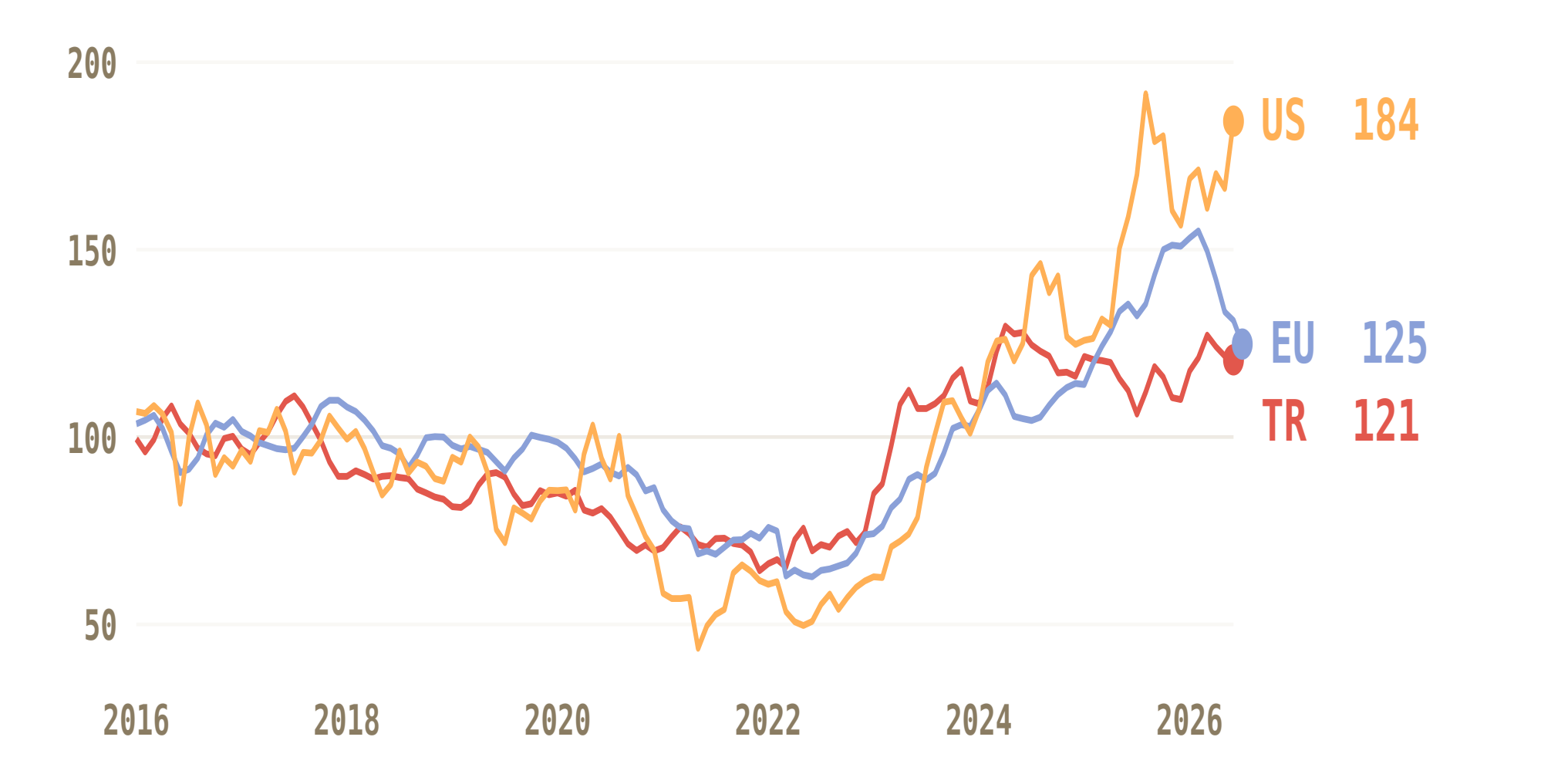
US cycles, 1971–2026 — the only record long enough to count them. Peaks and falls measured on the twelve-month average, so single-month spikes do not count as cycles.



TWELVE-MONTH AVERAGE • PLATEAU = MONTHS WITHIN 10% OF PEAK • FALL = PEAK TO THE NEXT LOW • A CYCLE PEAK IS A LOCAL MAXIMUM OVER ±18 MONTHS

Three markets. One cycle. At the same time.

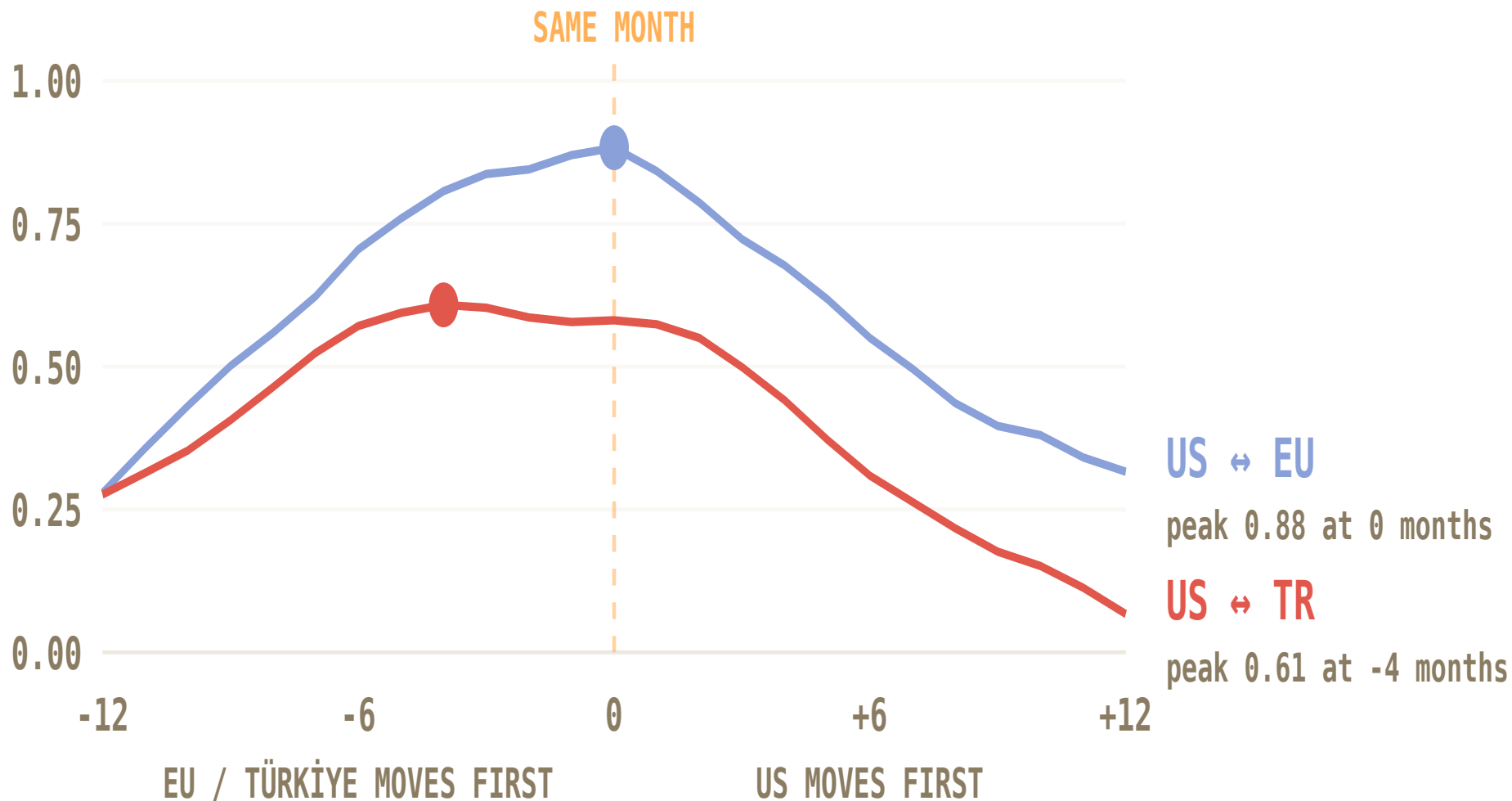
Now all three — each on its own 2016 = 100 base. Watch the shape, not the gap: levels are not comparable, timing is. The last year diverges — the US is still climbing while the EU cools — because each herd rebuilds on its own schedule.



EACH MARKET INDEXED TO ITS OWN 2016 AVERAGE = 100 · CHART STARTS 2016: FIRST YEAR ALL THREE SERIES EXIST · EU IS THE 27-MEMBER AGGREGATE, NOT ANY ONE COUNTRY

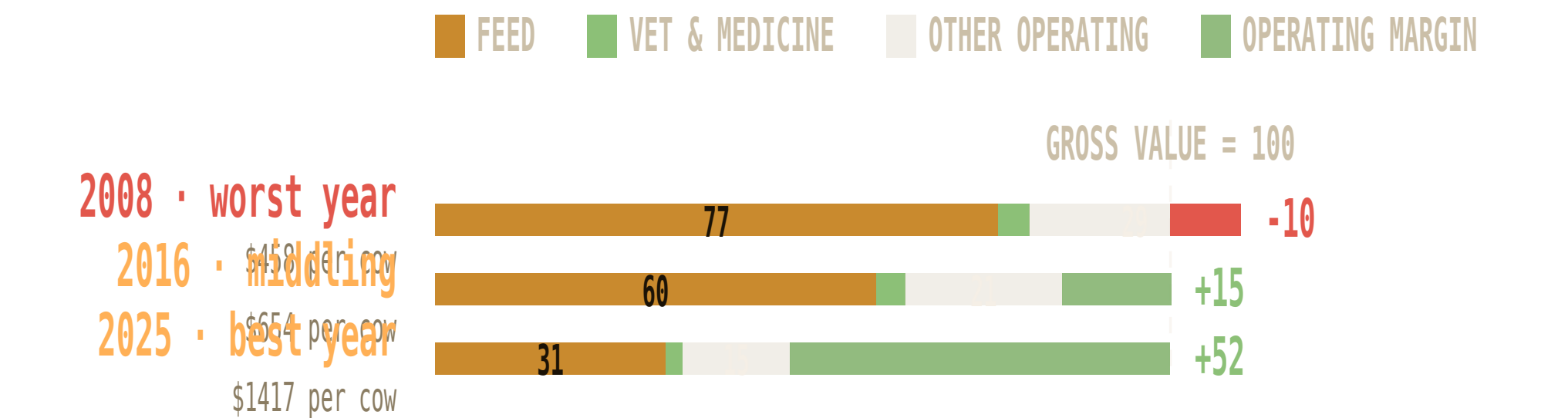
Yes — and nobody goes first, so there is no early warning.

Correlation of year-on-year change, tested at every shift from -12 to +12 months. A market that led would peak clearly off-centre. The EU peaks exactly at zero; Türkiye is flat across -4 to 0.



Feed was 77% of revenue. Now it is 31%.

USDA's own cow-calf accounts, per \$100 of gross value. Nothing modelled — these are surveyed costs. The feed line does the swinging; vet spend barely moves.



Operating margin, not profit — excludes overhead, unpaid family labour and capital recovery (USDA costs these at ~¾ of gross value). Cow-calf, not feedlot, so no calf-purchase line.

EVERY SOURCE IN THIS DECK

Meat & feed prices

US: BLS producer price indexes WPU0131 (slaughter cattle) ÷ WPU012202 (corn), via FRED, 1971 →
EU: European Commission beef reference price ÷ feed grain, 2015 →
Türkiye: TurkStat Yi-ÜFE TP.TUFE1YI.T17 ÷ T25, via CBRT EVDS, 2010 →

Cattle numbers

FAOSTAT cattle stocks (head), via Our World in Data, 1961–2024, all three markets

Veterinary spend

US: USDA ERS commodity costs and returns, cow-calf, 1996 →
EU: Eurostat Economic Accounts for Agriculture, item AM205000, 2005 →

Deflators

US: CPI-U (FRED CPIAUCSL) · Euro area: HICP (FRED CP0000EZ19M086NEST)

Breaks & limits

ERS changed survey basis in 2008 — no comparison spans it. EU veterinary line covers all livestock, not cattle alone. No per-head cost accounts published for Türkiye.

Six numbers. Each one checkable.

Each one reproducible from the committed data files. Three describe where we are; three describe what it does to us.

184

US margin at a 55-year record

2016 = 100. The twelve highest months in the whole series have all fallen since June 2025.

-34% / -33%

US and EU herds below their peaks

Scarcity is what makes the margin: parity and herd size correlate -0.47 in the same year.

0 of 5

previous US peaks that held

Every one reversed — 25 to 60%, over two to five years. Five observations: a pattern, not a law.

+0.70

US spend per head vs producer margin

The strongest relationship in the study. In the EU the same measure is flat — no cycle response at all.

0.88

US-EU correlation at zero lag

One global cycle with no leader and no follower, so there is no early-warning market to watch.

+4%

the real US pool, in seventeen years

Spend per head +15%, herd -9%. More per animal, fewer animals, and the two nearly cancel.

Three markets. Three different instructions.

Same cycle everywhere, three different consequences. The US is a timing problem. The EU is a volume problem. Türkiye is a visibility problem.

US

HOW MANY ANIMALS

-34%

vs peak · now 87m · bottom 2024

SPEND vs MARGIN

cyclical

r = 0.70

THE MONEY IN THE MARKET

+4%

17-yr real pool

TIME THE PRICE MOVE

2027 points up: the record margin still feeds spend per head, and the herd is bottoming. Phase the increase before the plateau breaks.

EU

HOW MANY ANIMALS

-33%

vs peak · now 72m · -1.4%/yr

SPEND vs MARGIN

flat

r = 0.00

THE MONEY IN THE MARKET

-5%

20-yr real pool

DEFEND THE VOLUME

2027 is not a bad year — it is the same year, ~1.5% smaller. No cycle upside, no cycle hit; the herd erodes regardless.

TÜRKİYE

HOW MANY ANIMALS

-6%

vs peak · now 17m · +2.4% 2024

SPEND vs MARGIN

no data

—

THE MONEY IN THE MARKET

n/a

not measurable

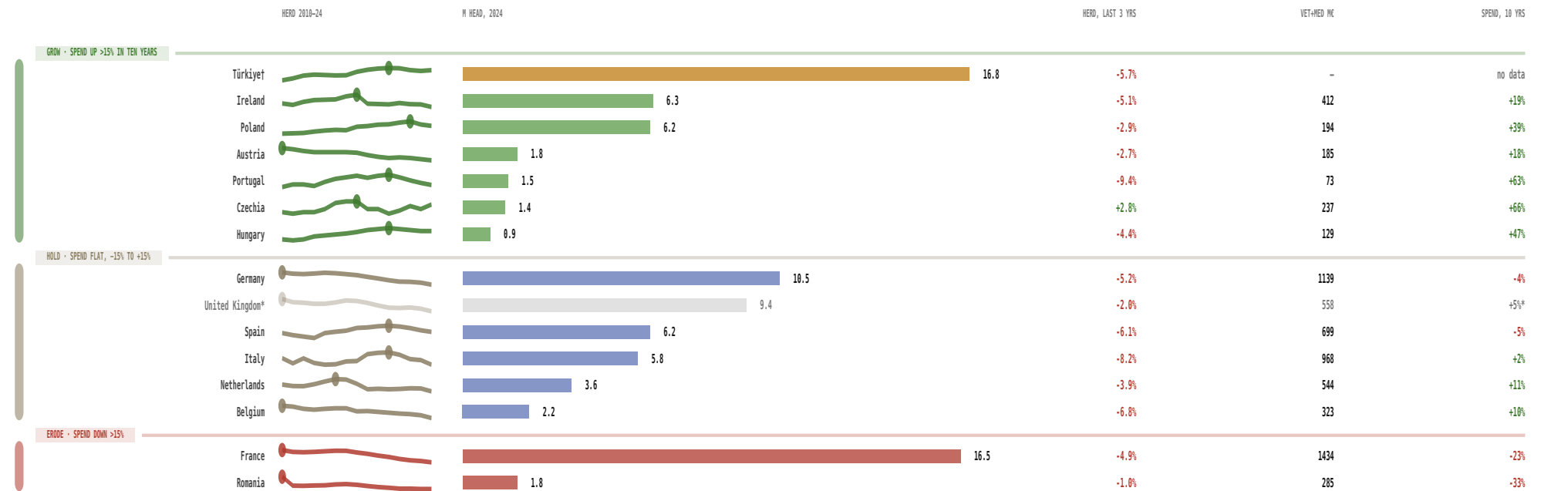
GROW, AND GET VISIBILITY

2027 is the only market where head count adds to volume — and the only one where we cannot see what producers spend.

What I am asking for: two to three weeks with the livestock commercial teams — map our cattle-segment sales in the CEEME markets through the last two downturns against this ratio, and rank that portfolio by what a producer defers first. **What I need from you is an introduction to the livestock lead.**

Every herd is shrinking. Only the spending separates them.

All cattle, per country. The line is each herd's own path since 2010 and the dot is its peak. Fourteen of the fifteen herds are smaller than three years ago — only Czechia grew. Hungary, Poland and Czechia did all their ten-year growth before 2020. There is no volume growth left to sell into in 2027. Clusters are grouped on real vet+med spend.



The 2027 price call. Every herd here is contracting, so the increase comes from price and from spend per head — not volume. **Grow** — Czechia, Portugal, Hungary, Poland, Ireland, Austria, Türkiye: real spend rose +18% to +66% through a decade of tighter herds. Take the full increase. **Hold** — Germany, Netherlands, Belgium, Italy, Spain: spend steady, herds not. Price carries the number; do not plan volume on top. **Erode** — France, Romania: spend down -23% and -33%. Defend share, do not lead on price. **United States (slides 4-7):** record margin absorbs price — front-load to January, keep 2028 flexible.

HERD: FAOSTAT, ALL CATTLE, DAIRY+BEEF, ANNUAL TO 2024 · SPARKLINE 2010-2024, EACH COUNTRY SCALED TO ITS OWN MIN AND MAX, DOT = PEAK YEAR · HERD LAST 3 YRS = 2021-2024, SAME SIGN ON A 5-YEAR WINDOW FOR ALL FIFTEEN · SPEND: EUROSTAT EAA "VETERINARY EXPENSES" = VET SERVICES + MEDICINES, ALL LIVESTOCK NOT CATTLE ALONE, 2025 LEVELS · 10-YR SPEND CHANGE IS REAL, DEFLATED BY EURO-AREA HICP (APPROX. FOR NON-EURO PL/RO/CZ/HU) · A 3-YEAR REAL SPEND CHANGE IS NOT SHOWN: THE 2022-23 INFLATION SPIKE MAKES IT FLIP SIGN WITH THE BASE YEAR · *UK SPEND SERIES ENDS 2020 · †NO SPEND SOURCE FOR TÜRKİYE · EU ROWS = 90% OF THE EU27 HERD · US IS ON A DIFFERENT SPEND SOURCE (USDA ERS, \$/COW) AND IS NOT COMPARABLE HERE